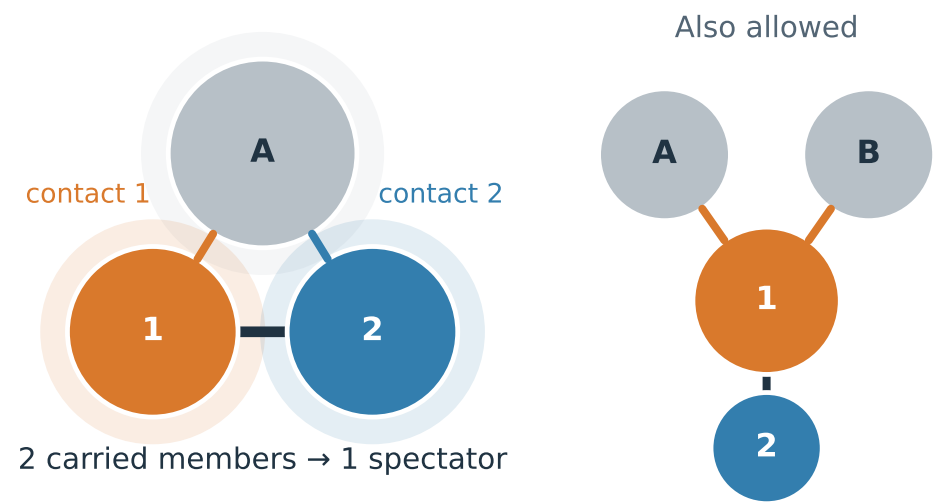


The two-contact update

Combine compatible docking preferences into one rigid-oligomer proposal; then test the full physical move.

1 Two interfaces, one rigid pose

Colored bodies move together; gray spectators stay fixed during this trial.

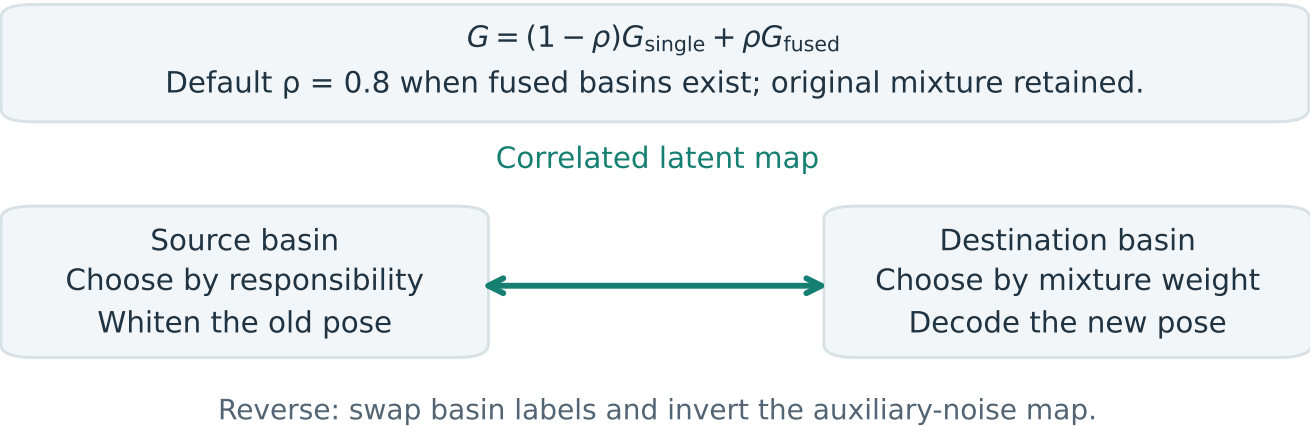


Each interface constrains the same collective pose H : $A_a^{-1} H U_i$

U_i fixes member i inside the oligomer. Halos represent depletant-exclusion regions.

3 Use the reversible basin map

Single-contact and fused basins share one normalized catalogue G .



Carry every selected member with the same rigid transformation.

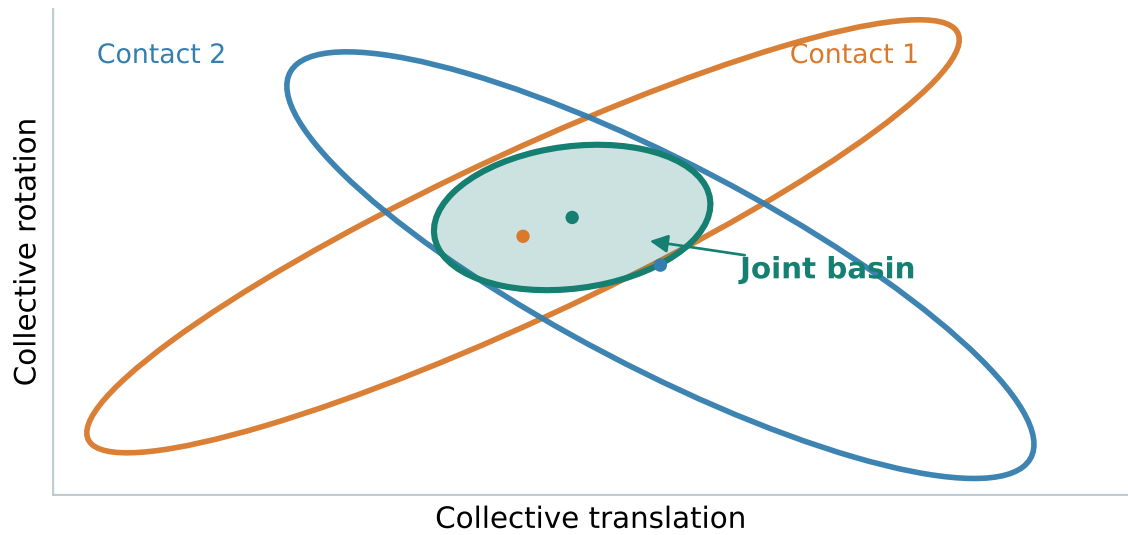
Source and destination can each be single or fused; keep the defensive uniform branch.

Same catalogue on reversal: fixed spectators + wall + rounded internal geometry; reject trials that change that key.

Schematic geometry and Gaussian slice only. Two distinct member-spectator interfaces guide the proposal; binding and native registry are not guaranteed.

2 Fuse compatible constraints

A two-dimensional slice of the six-dimensional collective pose space.



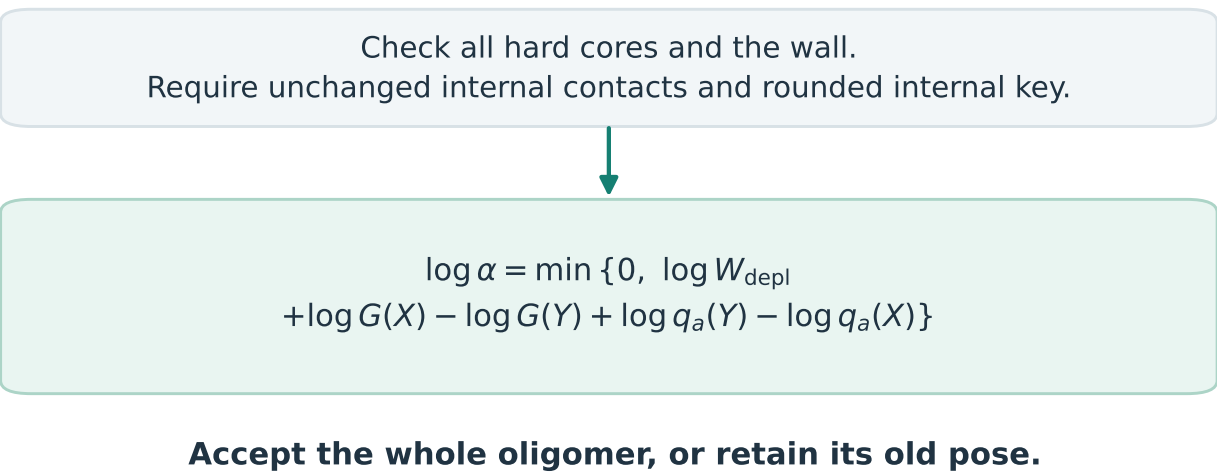
Fit: $H_* = \arg \min_H [\|z_1(H)\|^2 + \|z_2(H)\|^2]$

Local precision: $\Sigma_*^{-1} = J_1^T J_1 + J_2^T J_2$

The Gaussian approximates a proposal basin, not its physical statistical weight.

4 Check the complete physical move

A promising fitted center does not guarantee a valid or favorable trial.



W_{depl} : full many-body Poisson depletion factor; q_a : primary-anchor selection.

Use the full catalogue density G , including its pose-coordinate Jacobians.